Research Proposal

From Interpretation to Action: Distilling LLM Microstructure Priors into Low-Latency Position-Control Policies

LLM-to-RL Knowledge Distillation for Microstructure-Aware Short-Horizon Trading

Draft v0.5 (Pre-submission Final) — March 2026
Primary Target: ICAIF 2026

Abstract

We investigate whether Large Language Models (LLMs) can generate directional microstructure signals that, when distilled into lightweight neural networks and used to warm-start reinforcement learning, yield short-horizon position-control policies with improved sample efficiency and robustness. We propose MIND (Market INterpretation Distillation), a three-stage pipeline. In Stage 1, an LLM teacher generates directional scores from structured, position-independent limit order book (LOB) summaries; these signals are validated against supervised baselines on information coefficient (IC), sign stability, and regime-conditional performance—producing a standalone empirical result on LLM market-interpretation ability. In Stage 2, the teacher's scores are distilled into a compact student network. In Stage 3, the student's shared encoder and direction head initialize a PPO agent that jointly learns directional and position-sizing decisions under stylized trading frictions, with the sizing head learned from scratch via RL. We evaluate on four KRX equities with distinct liquidity and volatility profiles. Our claim is narrow: LLM-derived microstructure priors, distilled and used to warm-start RL-based position control, improve sample efficiency, robustness on held-out instruments and regimes, and low-latency deployment feasibility relative to cold-start RL and direct supervised baselines trained on identical features.

1. Introduction

Short-horizon directional trading under microstructure-aware constraints requires interpreting high-dimensional order book states and translating those interpretations into position adjustments within tight latency budgets. Existing approaches face a fundamental trade-off: hand-crafted rule-based strategies leverage known microstructure signals but are brittle to regime shifts [citation needed]; reinforcement learning agents learn flexible policies but suffer from high sample complexity and initialization sensitivity, particularly in non-stationary financial environments where exploration is costly [citation needed: RL sample inefficiency in financial trading].

Recent work has shown that LLMs, given structured numerical inputs with domain-specific prompts, can produce contextual reasoning about financial states [citation needed]. However, LLMs cannot serve as real-time trading policies: API latency (hundreds of milliseconds to seconds) exceeds typical signal lifetimes, and inference costs scale linearly with decision frequency. Moreover, LLMs lack the ability to perform state-dependent continuous optimization for position sizing—deciding how aggressively to adjust inventory given current spread, estimated impact, and inventory risk [citation needed: inventory risk and transaction cost modeling].

This gap motivates the central question: can LLM-derived market interpretations serve as useful

priors for training lightweight, locally deployable position-control policies? We ask whether an LLM's structured reasoning—even if imperfect—provides a better RL initialization than random weights or standard supervised pre-training.

Why an LLM Teacher?

The hypothesis is structural, not a claim of LLM superiority. LLMs can synthesize heterogeneous features (spread, imbalance, trade flow, price impact) via in-context reasoning without explicit feature-interaction engineering. Unlike a supervised MLP or gradient-boosted tree, an LLM can express conditional logic ("imbalance is positive but spread is widening, suggesting liquidity withdrawal") that captures interactions hard to specify a priori. Whether this translates into measurably higher IC is an empirical question that Stage 1 rigorously tests.

Scope and Claims

We constrain scope to microstructure-aware short-horizon trading, specifically RL-based position control under stylized trading frictions. The environment models proportional transaction fees, linear temporary market impact, and quadratic inventory penalties. It does not simulate queue position, partial fills, order book regeneration, or maker/taker dynamics. The action space governs position adjustment (direction and sizing fraction), not individual order placement. Claims pertain to sample-efficient RL initialization and robustness on held-out instruments with differing liquidity and volatility profiles, not to execution simulation fidelity or production deployment readiness.

Paper Structure

Stage 1 (Section 3.1) is a standalone signal validation: it evaluates whether the LLM teacher, given only market-state features (no inventory information), produces higher-quality directional signals than supervised baselines on identical features. Stage 2 (Section 3.2) distills the teacher's signal into a compact student whose architecture maps directly onto the PPO policy network. Stage 3 (Section 3.3) is the primary contribution: the distilled student warm-starts a PPO agent for position control under trading frictions, and we measure whether this initialization improves sample efficiency and final policy quality. The observation space in Stage 3 is strictly larger than in Stage 1: it includes inventory state (current position, unrealized P&L, time since last trade) that was deliberately excluded from the teacher's input.

2. Research Questions

RQ1. Teacher Signal Quality. Do LLM-generated directional scores carry higher IC and rank IC than supervised baselines (MLP, LightGBM, TCN) trained on the same market-state features and the same prediction target? Under what regimes does the LLM outperform or underperform?

RQ2. Distillation Fidelity. What fraction of the teacher's directional signal quality is preserved when distilled into a compact student? How does fidelity vary with student capacity and dataset size?

RQ3. Warm-Start Advantage. Does warm-starting PPO from a distilled student improve sample efficiency and final policy quality compared to cold-start PPO? Is the gain attributable to the LLM-specific prior, or would distillation from the best supervised baseline yield comparable results?

RQ4. Position-Sizing via RL. On the 2D action space (direction × sizing fraction), does RL learn state-dependent sizing behaviors that a directional-only teacher cannot provide?

RQ5. Deployment Feasibility. Does the final student policy achieve sub-millisecond model inference, and what is the cost reduction vs. the LLM teacher?

3. Proposed Framework: MIND

MIND (Market INterpretation Distillation) is a three-stage pipeline: **directional signal modeling** → **policy-prior distillation** → **RL-based position control**.

3.1 Stage 1: Teacher Signal Generation and Validation

Stage 1 evaluates the teacher's ability to interpret market states and generate directional signals. It is a standalone signal-quality experiment: no trading frictions, no inventory dynamics. The teacher receives only market-state features and predicts the direction and relative magnitude of short-horizon price moves.

Prediction Target

Definition. The prediction target is the raw forward mid-price return over horizon h:

```
y_t(h) = (mid_{t+h} - mid_t) / mid_t
```

where mid_t is the mid-price at step t and h is the prediction horizon in resampled time steps. We evaluate at $h \in \{1, 5, 10, 20\}$ steps (approximately 10s, 50s, 100s, 200s at 10s resampling).

Rationale for raw returns (no cost adjustment). Stage 1 evaluates market-interpretation ability: can the teacher predict where the mid-price is heading? Transaction costs, market impact, and inventory penalties are properties of the agent's execution and position-management policy, not of the market state being interpreted. Including costs in the prediction target would conflate interpretation quality with execution assumptions. Trading frictions enter

Stage 3, where the RL reward explicitly penalizes costs and inventory risk.

Label generation. Non-overlapping forward windows: at horizon h, the label at time t uses [t, t+h]; the next label is at t+h. When overlapping windows are used as a sensitivity check, Newey-West standard errors with lag h are applied.

Normalization. Returns y_t(h) are z-scored within each trading day to remove intraday drift.

Input Representation

Stage 1 input is market-state only—no inventory or position information. The teacher receives: mid-price, spread (bps), order imbalance, depth imbalance (L1), net trade flow, volume surprise, price impact estimates (buy/sell), and trailing mid-price returns (5/10/20 steps). Current position, unrealized P&L, and other inventory-related features are excluded.

Rationale. Stage 1 is a pure market-interpretation task. Including inventory state would make the teacher's signal inventory-conditional, conflating market prediction with position management. The student inherits the teacher's market-interpretation capability; inventory awareness is added in Stage 3 when the PPO observation space is augmented with inventory features that the encoder has not seen during distillation. This clean separation ensures the teacher is evaluated on market-state understanding alone.

LLM inputs are serialized as compact JSON with semantic labels. Raw LOB arrays are excluded [citation needed: LLM numerical reasoning limitations]. Text-narrative and hybrid variants are tested as sensitivity checks.

Output Schema

Field	Range	Role	Description
score	[−1, 1]	Primary	Directional signal; sole distillation target. Predicts sign and relative magnitude of y_t(h).
confidence	[0, 1]	Auxiliary	Self-assessed confidence; analysis only [citation needed: LLM confidence miscalibration]
expected_return_bps	float	Auxiliary	Magnitude estimate; analysis only
horizon_steps	int	Fixed	Set in prompt, not generated; same h as label
reason	string	Qualitative	Appendix analysis only; never enters training

Stage 1–Stage 3 Alignment

The teacher's score predicts sign(y_t(h))—the direction of the next mid-price move. In Stage 3,

the RL reward is: gross P&L from holding a position minus trading costs minus inventory penalty (defined in Section 3.3). The teacher's directional prior aligns with the gross P&L component: correct directional predictions lead to positive mark-to-market gains, which the RL agent then learns to extract net of frictions. The teacher provides a prior on "where" to trade; RL learns "how much" and "at what pace" under cost constraints.

Teacher Validation Protocol

The LLM teacher is compared against supervised baselines (MLP, LightGBM, TCN) trained on identical market-state features to predict the same $y_t(h)$. If the LLM does not outperform the best supervised baseline, the downstream pipeline uses the strongest available teacher.

Metric	Description
IC / Rank IC	Pearson and Spearman correlation between score and $y_t(h)$ at each h
IC Decay Profile	IC as a function of h; characterizes signal half-life
Directional Accuracy (full)	Hit rate of sign(score) vs. sign($y_t(h)$) over all samples
Directional Accuracy (|score| ≥ ε)	Same metric restricted to samples where |score| ≥ ε (default ε = 0.1); filters out weak/ambiguous signals
Top/Bottom Quantile Accuracy	Directional accuracy computed on the top and bottom 20% of |score|; tests whether strong-conviction signals are more accurate
Sign Stability	Fraction of consecutive steps where sign(score) is unchanged
Regime-Conditional IC	IC in high/low volatility, wide/narrow spread, trending/mean-reverting regimes
Prompt Sensitivity	Score variance across 3+ prompt variants on the same inputs
Comparative Advantage	Paired IC: LLM vs. best supervised baseline with bootstrap CIs

Neutral-signal handling. When $|\text{score}| < \varepsilon$ the teacher is effectively abstaining. Directional accuracy computed on the full sample may be misleading if many predictions are near zero. The thresholded accuracy ($|\text{score}| \geq \varepsilon$) and quantile accuracy metrics address this: they test whether the teacher's strong-conviction signals are directionally reliable, which is what matters for downstream distillation. We report $\varepsilon = 0.1$ as default and sweep $\varepsilon \in \{0.05, 0.1, 0.2\}$ in sensitivity analysis.

3.2 Stage 2: Knowledge Distillation (Policy Prior)

Stage 2 distills the teacher's directional signal into a compact network whose architecture maps directly onto the PPO policy's shared encoder and direction head.

Teacher Dataset

The best-performing teacher (LLM or strongest supervised baseline, determined in Stage 1) is

run across the training data, generating (market_state_features, score) pairs. No inventory features are included. Target: 50K–200K samples.

Student Architecture

The student mirrors the PPO encoder + direction head:

- Shared encoder: 3 hidden layers (256–128–64), ReLU, dropout 0.1. Input: market-state features only (same as Stage 1).
- Score head: single linear output → tanh, predicting the teacher's score ∈ [−1, 1].

Distillation Objective

Primary loss: MSE on score with directional regularization:

```
L = MSE(s_student, s_teacher) + λ · max(0, −sign(s_teacher) · s_student)
```

λ is tuned on a held-out validation set. Confidence is not a distillation target.

Score-to-Direction Transfer Rationale

Why regress a continuous score rather than classify into 3 directions? The teacher's continuous score encodes not only direction but also conviction magnitude: a score of +0.8 carries more information than +0.1. Regression preserves this ordinal structure; a 3-class classification label would discard magnitude and treat all buys identically. During transfer to the PPO direction head (Section 3.3), the continuous score is mapped to 3-class logits via a thin adapter layer. This mapping incurs some information loss, but the shared encoder's weights—which capture the feature representations that produce the continuous score—transfer losslessly.

Ablation variant. We include a sensitivity experiment that adds an auxiliary 3-class cross-entropy loss (sign-based: sell/flat/buy with a dead-zone $|\text{score}| < \varepsilon$ mapped to flat) alongside the primary MSE loss during distillation. This tests whether explicitly training toward directional categories during distillation improves the warm-start transfer, or whether the regression-only path is sufficient.

Supervised-Prior Control

A critical control replaces teacher labels with predictions from the best-performing supervised baseline. If PPO warm-started from this supervised-distilled student matches the LLM-distilled variant, the LLM's unique contribution is limited [citation needed: teacher-student distillation and warm-start in RL].

3.3 Stage 3: RL-Based Position Control

Stage 3 is the primary contribution. It fine-tunes the distilled student as a PPO policy for position control under trading frictions, adding: (a) optimization against a friction-aware reward, and (b) state-dependent position-sizing decisions.

PPO Architecture and Initialization

Architecture (shared encoder, dual action heads, value head):

- Shared encoder: same 3-layer MLP as the distilled student (256–128–64). Input: market-state features + inventory state (q_t, unrealized P&L, steps since last trade, remaining budget). Note: the encoder input dimension in Stage 3 is larger than in Stage 2 because inventory features are appended. The additional input dimensions are initialized with small random weights in the encoder's first layer; all other encoder weights are copied from the student.
- Direction head: 3-class softmax over {−1, 0, +1}. Initialized from the student's score head via a thin adapter: the student's tanh output is mapped to logits such that positive scores bias the +1 class, negative scores bias −1, and near-zero scores bias 0. The adapter is a single linear layer with analytically set initial weights (no additional training).
- Sizing head: 5-class softmax over {0.0, 0.25, 0.5, 0.75, 1.0}. Initialized to near-uniform logits. No pre-trained prior; learned entirely through RL.
- Value head: separate MLP branch from encoder; initialized randomly (standard PPO).

What the warm-start transfers. Two components: (1) a shared representation (encoder weights trained to extract directionally informative features from market-state data), and (2) a directional prior (direction head biased toward the teacher's signal). The sizing head starts neutral; the value head starts random. This asymmetric initialization supports ablation: encoder-only transfer (re-initialize both action heads) vs. encoder+direction transfer vs. full architecture.

Cold-start control. Same architecture, all components initialized with Xavier/He random weights. Identical parameter count.

Action Space

Symbol	Space	Definition
d_t	{−1, 0, +1}	Target position direction: short, flat, long
f_t	{0.0, 0.25, 0.5, 0.75, 1.0}	Fraction of gap between current and target position to close this step

Position dynamics. Q_{max} = maximum allowed inventory. Target: $q^*_t = d_t \cdot Q_{max}$. Adjustment:

$$\Delta q_t = f_t \cdot (q^*_t - q_t) \qquad q_{t+1} = q_t + \Delta q_t$$

This is an inventory-adjustment scheme, not an order-placement policy.

Reward Definition

Three components, combined into the per-step training reward:

Gross P&L (mark-to-market, before any frictions):

$$r_{pnl,t} = q_t \cdot (mid_{t+1} - mid_t)$$

Trading cost (proportional fee + linear temporary impact):

$$c_{trade,t} = c_{prop} \cdot |\Delta q_t| \cdot mid_t + \alpha \cdot |\Delta q_t| \cdot spread_t$$

c_{prop} calibrated to KRX fee schedule; α scales linear temporary impact [citation needed: Almgren & Chriss, 2001].

Inventory penalty:

$$c_{inv,t} = \phi \cdot q_t^2$$

Penalizes large positions [citation needed: inventory risk penalty in RL trading].

Net reward (the RL training signal):

$$r_t = r_{pnl,t} - c_{trade,t} - c_{inv,t}$$

Terminology convention. “Gross P&L” = $r_{pnl,t}$ (before frictions). “Trading cost” = $c_{trade,t}$. “Inventory penalty” = $c_{inv,t}$. “Net reward” = r_t. These four terms are used consistently throughout the paper. We do not use “implementation shortfall” because that term (Perold, 1988) refers specifically to cost vs. decision-price benchmark in execution contexts, which does not match our combined directional P&L + friction + inventory formulation.

Terminal Forced Liquidation

At the end of each episode (end of trading-day window), any remaining position is forcibly closed. The terminal step accounting is as follows:

- The forced close sets $d_T = 0$ (target flat) and $f_T = 1.0$ (full liquidation), producing $\Delta q_T = -q_T$.
- Gross P&L for the terminal step: $r_{pnl,T} = q_T \cdot (mid_{T+1} - mid_T)$, using the next-step mid-price. If the episode ends at the last available step, $mid_{T+1} = mid_T$ (no overnight gap is modeled).
- Trading cost for forced close: $c_{trade,T} = c_{prop} \cdot |q_T| \cdot mid_T + \alpha \cdot |q_T| \cdot spread_T$. The full spread and impact cost of unwinding is charged.
- Inventory penalty at terminal step: $c_{inv,T} = \varphi \cdot q_T^2$, computed on the pre-close position. After close, $q_{T+1} = 0$.

- The terminal net reward r_T follows the same formula: $r_{pnl,T} - c_{trade,T} - c_{inv,T}$. No special terminal bonus or penalty is added beyond the standard components.

Evaluation accounting. Cumulative gross P&L, cumulative trading cost, cumulative inventory penalty, and cumulative net reward all include the terminal step. There is no double-counting: the inventory penalty at time T penalizes holding into the close, and the trading cost at T charges the liquidation friction. These are complementary, not redundant—the penalty discourages arriving at T with large inventory, while the trading cost prices the actual liquidation.

Environment

- Observations: market-state features (same as Stage 1) + inventory state (q_t, unrealized P&L, steps since last trade, remaining budget).
- Episodes: rolling fixed-length windows (one trading day); no train/val/test overlap.
- Terminal: end of window; forced close as described above.

Realism boundaries. No queue position, partial fills, or order book regeneration. Impact is linear temporary [citation needed: concave impact functions]. These are scope constraints: we evaluate policy initialization under stylized frictions, not execution simulation fidelity.

4. Experimental Design

4.1 Data and Instruments

KRX LOB snapshots via KIS Open API, resampled at 10s (primary), 5s/30s (sensitivity).

Instrument	Ticker	Sector	Liquidity	Volatility	Role
Samsung Elec.	005930	Semicon	High	Low–Med	Train
SK Hynix	000660	Semicon	High	Med–High	Train
NAVER	035420	Internet	Medium	Medium	Held-out
KakaoBank	323410	Finance	Low–Med	High	Held-out

Samsung and SK Hynix serve as training instruments (high liquidity, same sector, different volatility). NAVER and KakaoBank serve as held-out instruments with differing sector, liquidity, and volatility profiles. We report performance on held-out instruments as a robustness check, not as evidence of broad cross-market generalization—all four instruments are KRX-listed Korean equities.

Temporal Split

Strictly chronological expanding-window walk-forward. Training grows; validation is a fixed buffer; test is always OOS. No temporal leakage. Cross-period robustness: results on ≥3 non-overlapping test windows. Cross-instrument check: train on Samsung + SK Hynix, evaluate transfer to NAVER and KakaoBank.

4.2 Baselines

Baselines are organized hierarchically. Three primary comparison axes follow the tables.

Tier 1 — Heuristic

Baseline	Description	Controls For
Rule-based Micro	Imbalance + spread threshold; fixed sizing	Feature value without learning

Tier 2 — Direct Supervised

Baseline	Description	Controls For
Supervised MLP	MLP predicting y_t(h) on same features	Neural supervised
LightGBM	Gradient-boosted tree on same features	Non-neural supervised
TCN	Temporal convolution on feature windows	Sequence baseline

Tier 3 — Distilled (no RL)

Baseline	Description	Controls For
Distilled, no RL	Student after imitation learning; heuristic sizing	Distillation alone

Tier 4 — RL Policies

Baseline	Description	Controls For
PPO Cold-start	Random init; same env and 2D action space	RL without prior
Best-Sup.-Dist. + PPO	PPO warm-started from best supervised distillation	Generic supervised warm-start
MIND (LLM-Dist. + PPO)	PPO warm-started from LLM-distilled student (proposed)	Full pipeline

Auxiliary

Baseline	Description	Controls For
LLM Direct	LLM at inference time; uniform sizing	Stress-test only

Primary comparison axes:

1. MIND vs. PPO Cold-start: does teacher-distilled initialization improve sample efficiency and final policy quality?
2. MIND vs. Best-Supervised-Distilled + PPO: is the LLM prior specifically valuable, or would any supervised warm-start suffice?
3. MIND vs. Distilled (no RL): does RL refinement under frictions add value beyond distillation alone?

The LLM Direct Policy is reported as an auxiliary data point for cost-performance context; it is not a primary comparison target.

4.3 Evaluation Metrics

All terms follow the definitions in Section 3.3.

Policy Performance (Stage 3)

- Gross P&L: cumulative $r_{pnl,t}$ (bps). Reported separately from frictions for decomposition.
- Trading cost: cumulative $c_{trade,t}$ (bps), decomposed into fee and impact components.
- Inventory penalty: cumulative $c_{inv,t}$.
- Net reward: cumulative r_t = gross P&L − trading cost − inventory penalty (bps). Primary metric.
- Sharpe ratio: annualized, on per-episode net reward.
- Maximum drawdown: peak-to-trough in cumulative net reward.
- Turnover: $\Sigma|\Delta q_t| / (T \cdot Q_{max})$.

Position Control Quality

- Sizing fraction distribution: f_t histogram conditioned on spread width and volatility regime.
- Inventory trajectory variance: $\mathrm{Var}(\Delta q_t)$.
- Average trade size: mean $|\Delta q_t|$ per active step.

Teacher Signal Quality (Stage 1)

- IC, rank IC at $h \in \{1, 5, 10, 20\}$; directional accuracy (full and thresholded); top/bottom quantile accuracy; sign stability; regime-conditional IC; prompt sensitivity.

Distillation Fidelity (Stage 2)

- Score MSE, directional agreement rate, Spearman ρ.

Operational

- Inference latency (mean, p95, p99) [citation needed: practical inference benchmarking].
- LLM API cost per episode; malformed response rate.

4.4 Statistical Rigor

Stratified bootstrap CIs (10K resamples). Harvey-Liu-Zhu correction (t > 3.0) across strategy variants [citation needed: Harvey, Liu & Zhu, 2016]. Walk-forward validation. RL: 5 seeds; rliable IQM, P(improvement), bootstrap CIs [citation needed: Agarwal et al., 2021].

5. Sensitivity and Ablation Plan

Dimension	Variants	Purpose
Resampling	5s, 10s, 30s	Signal decay vs. latency
Input Format	JSON, narrative, hybrid	LLM input sensitivity
Feature Subset	Full, spread+imb., minimal	Feature importance
LLM Model	GPT-4o, Claude Sonnet, Gemini	Model-dependent quality
Teacher Source	LLM, best supervised	LLM prior uniqueness
Student Architecture	MLP-256, MLP-128, TCN	Capacity vs. fidelity
Distillation Loss	MSE-only, MSE+sign-class aux	Regression vs. classification transfer
Distillation Size	10K, 50K, 100K, 200K	Data efficiency
PPO Budget	100K, 500K, 1M, 2M steps	Convergence advantage
Action Space	1D direction, 2D dir+sizing	Joint vs. decoupled
Warm-start Ablation	Encoder-only, enc.+dir., full	Representation vs. prior
Neutral Threshold ε	0.05, 0.1, 0.2	Signal filtering sensitivity
Instrument	Samsung, Hynix, NAVER, KakaoBank	Held-out robustness

6. Expected Contributions

Central claim: LLM-derived microstructure priors, distilled into a lightweight student and used to warm-start RL-based position control, improve sample efficiency, robustness on held-out instruments and regimes, and low-latency deployment feasibility relative to cold-start RL and direct supervised baselines under stylized trading frictions.

1. Empirical assessment of LLM microstructure interpretation: comparison of LLM teacher signals against supervised baselines on IC, rank IC, thresholded/quantile directional accuracy, regime-conditional performance, sign stability, and prompt sensitivity. Stage 1 is a standalone contribution on LLM market-state interpretation ability.
2. MIND pipeline with controlled ablations: three-stage framework (directional signal → policy prior → RL position control) with comparison between LLM-distilled and best-supervised-distilled warm-starts, separating LLM prior value from generic supervised pre-training benefit.
3. Warm-start decomposition: ablation isolating representation transfer (encoder) from directional prior transfer (direction head), plus regression-only vs. regression+classification distillation.
4. Deployment feasibility: latency profiling showing sub-millisecond student inference vs. LLM teacher latency and cost.

7. Limitations and Risks

Risk	Impact	Mitigation
LLM teacher ≤ best supervised	LLM narrative weakens	Valid negative result; pipeline uses best teacher
Distillation loss too high	Student underperforms	Ablate architecture, data, loss
Warm-start vanishes with more RL	Only sample efficiency	Report learning curves; efficiency matters
Sup.-distilled PPO matches MIND	LLM prior redundant	Report transparently; pipeline contributes
Env. simplifications	Realism skepticism	“Position control under stylized frictions”
KRX-only, 4 instruments	Limited external validity	Held-out robustness check, not broad generalization claim
API cost	Small teacher dataset	30s resampling; cache; cheaper models

Acknowledged Simplifications

- No queue, partial fills, OB regeneration; linear temporary impact.
- LOB snapshots (KIS API), not full message-feed.
- Action = position sizing, not order placement.
- Single exchange (KRX); held-out instrument robustness, not cross-market generalization.

8. Timeline

Phase	Tasks	Deliverable
M1–2	Data pipeline (4 instruments), serializer, LLM client, prompts, supervised baselines, teacher validation w/ thresholded metrics	Stage 1: signal quality report
M2–3	Teacher dataset, student arch, distillation (MSE + classification ablation), sup.-prior control	Stage 2: distilled student
M3–4	PPO (warm/cold), 2D actions, warm-start ablation, learning curves, held-out instrument eval	Stage 3: full MIND + ablations
M4–5	Sensitivity, robustness, latency, stats, terminal accounting validation, writing	Paper draft

9. Target Venue

Primary: ICAIF 2026. Microstructure + RL + deployment feasibility aligns with ICAIF.

Secondary: NeurIPS 2026 Workshop on Foundation Models for Decision Making.

Positioned at financial microstructure × knowledge distillation × sample-efficient RL.

10. Infrastructure

Integrates with proj_rl_agent: AlphaTradingEnv (2D action, terminal liquidation), feature pipeline, PipelineRunner, ExperimentTracker. New: LLM client, serializer, distillation trainer, supervised baselines.

POMDP Belief module as additional reference: Belief-augmented cold-start PPO tests LLM priors vs. hand-designed latent representations.

Appendix: Revision Log (v0.4 → v0.5)

Issue	v0.4 Weakness	v0.5 Fix
Stage 1 input: position	current position included in Stage 1 input, blurring market-interpretation claim	Removed position from Stage 1 input; added explicit rationale and observation-space expansion note for Stage 3
Score→direction transfer	tanh→sign mapping given without justification	Added rationale (regression preserves magnitude); added

		auxiliary classification loss ablation variant
Directional accuracy	Full-sample sign accuracy treats weak and strong signals equally	Added thresholded accuracy (|score| ≥ ε), top/bottom quantile accuracy, neutral-handling explanation, ε sweep
Cross-instrument claim	Phrasing could imply broad generalization from 4 KRX stocks	Scoped to “held-out instruments with differing profiles”; acknowledged KRX-only limitation explicitly
Terminal accounting	Forced close mentioned but accounting unspecified	Full terminal liquidation section: price basis, cost charging, inventory penalty timing, no double-counting argument
Terminology	Mostly clean but residual risk of mixing terms	Final pass: gross P&L / trading cost / inventory penalty / net reward used consistently; IS banned with rationale
Baseline clarity	Comparison axes stated but could be sharper	Numbered primary axes 1-2-3 after tables; LLM Direct explicitly demoted to auxiliary
Citations	Strategic but review pass needed	8 strategic placements confirmed: RL inefficiency, inventory risk, A-C, confidence miscalib., distillation, HLZ, rliable, concave impact, inference benchmarking

Remaining Residual Risks

- LLM teacher may not outperform supervised baselines—pipeline degrades gracefully but headline weakens.
- Linear temporary impact is a simplification; nonlinear impact could shift results.
- KRX-only with 4 instruments; cross-exchange transfer not tested.
- 5-bin discrete sizing is coarse; continuous Beta head is a natural extension.
- Inventory features appended to the encoder at Stage 3 were not seen during Stage 2 distillation; the encoder must adapt to these new inputs during RL. If this adaptation is slow, the warm-start advantage may be smaller than hypothesized.